

# THE PROBABILISTIC AUTHORITY PARADOX

Algorithmic management, Uber's evolving platform, psychological welfare, mental health AI, and the limits of machine judgment

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**Central thesis: The principal governance problem is not the existence of probability. It is the delegation of consequential authority without commensurate validation, transparency, human accountability and mechanisms for redress.**

Prepared for research and discussion, not as a claim of verified misconduct by any named platform, a clinical guideline or an independently peer reviewed empirical finding.

# Abstract

Organizations increasingly assign functions of supervision, compensation, scheduling, prioritization and decision support to algorithmic systems. This transition is often justified by scale, consistency and predictive performance. However, a prediction can be statistically useful and still optimize an institutional proxy that is poorly aligned with human welfare. This paper develops the proposed Probabilistic Authority Paradox: the risk that delegated decision authority expands faster than the organizational capacity to inspect, contest and correct automated decisions. Uber provides a longitudinal field research setting for studying compensation, information asymmetry and worker responses. Healthcare provides a high consequence comparison in which medical need can diverge sharply from operational proxies. Mental health requires two distinct inquiries: psychosocial effects of algorithmic control over work, and the risks of employing conversational AI as an apparent source of psychological care. The paper presents a proposed GRI research architecture, operational definitions, competing hypotheses, ethical safeguards and a governance model anchored in primary institutional and peer reviewed literature [1-9].

## 1. Starting point: what probabilistic systems can and cannot establish

The recent GRI sports prediction discussions serve as a useful illustration of miscalibrated confidence, not as scientific proof of model unreliability. A losing series of predictions cannot establish a systematic defect without a sufficiently large prospective sample, advance recording of stated probabilities, comparable events and a suitable scoring rule. Even a well calibrated 60 percent prediction fails approximately 40 percent of the time. No forecasting system guarantees individual outcomes merely because it expresses confidence.

Large language models generally produce text through probabilistic token selection conditioned on context, while the deployed system may also include retrieval, code execution, rules, and external models. A high probability for an output sequence is not equivalent to a calibrated probability that a proposition is true. Machine learning in transportation and healthcare also involves methods other than language models, including conventional statistical prediction, operations research and deterministic decision rules. It would therefore be inaccurate to conclude that every automated management decision is based solely on LLM token probabilities [2, 4].

The institutional question is more consequential: when uncertainty is translated into an offer, wage, warning, clinical recommendation or workflow action, who establishes the acceptable error rate? Who can see the underlying objective? Who can appeal a decision? Which errors are detectable, and how quickly can they be reversed?

## 2. Corporate management is more than rules and procedures

Classical corporate management incorporates operating procedures, legal obligations, performance targets and standardized evaluation. Yet actual managerial practice also involves tacit contextual knowledge, trust, discretion, conflict resolution, negotiation, developmental feedback and duties of care. A written policy can require respectful treatment or equitable escalation; it cannot exhaust the realities of a worker who is injured, financially distressed, grieving, exhausted or facing a mental health crisis.

Computational systems can enforce rules consistently and may outperform people on some narrow prediction tasks. However, formal consistency is not the same as substantive justice. Two workers can receive identical procedure based treatment despite facing materially different constraints, or different treatment that is rationalized by a black box score but cannot be meaningfully contested. The inability to perform human judgment should be specified rather than metaphysically asserted: existing systems may simulate empathic language, predict conditions, and escalate based on thresholds; they do not thereby assume professional obligations, embodied understanding, or legal and moral accountability.

### Five dimensions of accountable human management

- Context: interpretations of performance must account for conditions, constraints and competing duties.
- Explanation: affected individuals should understand the materially relevant reasons for consequential decisions.
- Discretion: policies should allow justified exceptions, proportionality and individualized accommodations.
- Recourse: errors require appeal paths, accountable ownership and documented correction.
- Care: management should recognize psychosocial risks, not just productivity indicators.

The International Labour Organization defines algorithmic management as tracked data used to organize, assign, monitor, supervise and evaluate work. Such systems may include machine learning, but they can also rely on conventional rules. The ILO's analysis of logistics and healthcare identifies efficiency benefits alongside challenges for job quality and intrusive surveillance [5, 6].

## 3. Uber as an evolving algorithmic management institution

Uber's driver application combines a marketplace, dispatch environment, performance measurement interface and pay communication system. The platform's published disclosures should not be mistaken for complete visibility into proprietary algorithms. Nonetheless, public materials establish changes important to research. In 2022, Uber described expansion of Upfront Fares and Trip Radar, showing drivers more information about expected payment and destination before acceptance [7]. Its September 2022 explanation states upfront compensation could depend on factors beyond fixed time and distance, including destination demand and pickup conditions [8].

This shift matters because transparency has at least three meanings: outcome transparency (the driver sees a proposed payment), process transparency (the driver understands the factors and relative weights producing it), and contestability (the driver can challenge a malfunction or materially misleading application). More of the first does not automatically produce the other two. Rosenblat and Stark’s empirical case study identified information and power asymmetries as features of Uber’s earlier platform management [1]. New interfaces may change the balance, but the question remains empirically open.

## Management functions performed or mediated by a platform

| Conventional function | Algorithmic platform operation               | Potential governance question            |
|-----------------------|----------------------------------------------|------------------------------------------|
| Scheduling            | Supply forecasts, incentives and positioning | Who bears idle time?                     |
| Assignment            | Dispatch and matching                        | Can workers evaluate selection fairness? |
| Compensation          | Real time trip offers and incentives         | What objectives shape pay?               |
| Evaluation            | Ratings and behavioral metrics               | Is there adequate recourse?              |
| Risk controls         | Identity, safety and fraud checks            | Who verifies false positives?            |

No public evidence cited here establishes a secret, intentionally coded \$20 hourly earnings cap. Establishing the precise causes of any earnings clustering requires controlled measurement and testing of alternative explanations.

## 4. The GRI approximately \$20 gross hourly band hypothesis

Prior GRI field notes report observations around \$20 gross per hour, alongside attributed late September 2026 Miami third party figures of \$18.78 for Uber and \$16.73 for Lyft. Those values should be treated as preliminary research leads until source snapshots, sampling criteria, definitions of hourly earnings and market conditions are independently archived. They are not pooled observations in a validated dataset, and they are not interchangeable estimates of net wages.

The central proposition is that a dynamic marketplace can produce an apparent earnings band while individual offers differ materially. At least five competing mechanisms warrant examination: supply and demand equilibrium; optimization of rider prices and driver incentives; driver entry, exit and repositioning; selection effects from observed completed rides; and the possibility of driver specific pricing differences. Several may operate simultaneously. Finding clustering would not identify the internal objective of a proprietary system.

## Measurement definitions and denominators

- Gross receipts per online hour: total platform receipts divided by all time connected and available to receive work.
- Gross receipts per engaged hour: receipts divided only by time allocated to active work; this may materially overstate economic return when waiting is substantial.
- Net operating return: receipts less appropriately allocated operating expenses divided by total online hours.
- Net economic return: net operating return adjusted for longer run vehicle capital cost, depreciation, insurance and other relevant costs without double counting.

As a sensitivity illustration rather than an empirical GRI finding, \$20 per online hour with \$7 allocated operating costs produces \$13 net operating return per hour before taxes and any omitted costs. The calculation demonstrates the importance of denominators and cost assumptions, not the actual net earnings of a particular driver.

## 5. Bias mechanisms beyond intentional discrimination

### 5.1 Objective function and proxy bias

An optimization problem requires a target. An algorithm designed to increase completed trips may deliver excellent completion statistics without maximizing driver welfare. A hospital model designed to forecast spending can accurately predict costs while failing to identify unmet medical need. The chosen target embeds managerial values. A mismatch between target and legitimate institutional purpose creates harm even when the numerical prediction is technically accurate.

### 5.2 Historical and representation bias

Training records reflect previous policies, geographic inequalities, consumer behavior, access to services and institutional decisions. Data may overrepresent frequent app users, connected populations or people whose outcomes are easy to measure. Underrepresented groups can receive weaker predictions even without any explicit protected attribute as an input.

### 5.3 Behavioral feedback and adaptive systems

Drivers react to payment offers through acceptance, rejection, time choice and repositioning. Their reactions affect demand fulfillment, observed trip data and future marketplace conditions. Apparent driver preferences may therefore be partly shaped by earlier platform decisions. In healthcare, resource allocation can influence utilization, which then feeds later spending based risk scores. Feedback makes cause and effect difficult to identify with simple cross sectional correlations.

## 5.4 Measurement and surveillance bias

Platforms may measure ratings, cancellations and time online more easily than exhaustion, anxiety, financial insecurity or difficult passenger interactions. A management system can optimize available data while missing psychological harm. Surveillance can also change conduct itself: workers adapt to measurable targets, sometimes at the expense of unmeasured quality.

## 5.5 Automation, explanation and authority bias

Users may interpret polished interfaces and fluent text as evidence of competence. Staff may defer to risk scores without adequate independent scrutiny, especially under time pressure. Automated decisions also create an attribution gap: a worker may be directed from a call center to the app, while frontline personnel have limited authority to override the system. The organization retains responsibility even if the immediate decision is generated by software.

# 6. Mental health pathway I: algorithmic management and worker welfare

The American mental health question is not restricted to chatbots offering therapy. It includes mental health consequences of work organized by surveillance, dynamic ratings, uncertain pay and mediated access to employment. The relevant research constructs are perceived autonomy, job insecurity, procedural justice, emotional exhaustion, sleep disruption, perceived surveillance, financial strain and perceived access to human support. These are measurable constructs, not assumptions that every driver develops a psychiatric condition.

## Mechanisms to investigate, not presumed causal findings

- Income volatility: inconsistent cash flow may increase perceived financial insecurity, especially when fixed expenses remain stable.
- Information asymmetry: uncertainty about why an offer, restriction or deactivation occurred can impair perceived procedural justice.
- Continuous evaluation: ratings and behavioral scores may incentivize emotional labor and discourage reporting of problematic interactions.
- Fragmented social support: remote app mediated management may reduce access to a familiar supervisor who can interpret contextual circumstances.
- Loss of control: workers may formally choose hours while experiencing strong economic pressure to remain available during algorithmically indicated periods.

A rigorous GRI study must avoid diagnosing individuals from app usage or work performance. Voluntary, consent based psychological questionnaires could be administered only with appropriate privacy protections and independent ethics review where applicable. Interviews can document subjective experiences, but correlations between algorithmic management and reported distress cannot be

interpreted automatically as causal effects. Confounders include preexisting financial pressures, health status, family circumstances, vehicle failures, housing costs and local market changes.

### **Managerial duty and an accountability gap**

When a traditional supervisor recognizes an employee in distress, the organization may possess established routes for accommodations, referrals and escalating risks. An automated dispatch or performance system may possess none of the contextual awareness needed to recognize that a deviation from expected behavior is connected to a crisis. It may instead record a cancellation, low availability, delayed response or unusual behavior as a performance event. This is a governance risk, not proof that Uber intentionally penalizes mental illness.

## **7. Mental health pathway II: the AI model as apparent counselor**

A distinct concern arises when people use generative AI for emotional support or psychotherapy like interactions. The American Psychological Association's November 2025 health advisory states that consumer chatbots and wellness applications are not substitutes for qualified mental health care, may lack scientific validation and can produce dangerous or inappropriate advice. It emphasizes insufficient crisis capabilities, possible sycophancy, cultural bias and the risk of encouraging unhealthy dependence [9]. The World Health Organization likewise flags inaccurate or biased outputs and automation bias in health settings [3].

### **Why conversational plausibility is not clinical adequacy**

A chatbot can reproduce language associated with compassion, but verbal fluency does not establish diagnostic reliability, therapeutic alliance, responsibility for a patient or the capacity to interpret nonverbal signals and longitudinal clinical history. A system that reflexively affirms a distressed user may amplify inaccurate self assessments or maladaptive beliefs. Conversely, an overly rigid intervention can dismiss context, confuse distress with imminent danger or erode trust. Neither agreeable responses nor highly precautionary responses can be assumed to constitute safe psychological care.

Clinical safety also involves escalation and continuity. A human clinician may evaluate risk, coordinate with family or emergency services when appropriate, and accept professional accountability. A general purpose AI system cannot be assumed to do those things reliably. The correct use of AI in mental health must be defined by intended purpose, validation evidence, clinician oversight and safeguards, not by the sophistication of the conversation alone.

## **The ambiguity of helpfulness**

A person may report feeling better immediately after a conversation yet experience no sustained improvement in clinical outcomes. Consumer satisfaction, session length or positive sentiment are not adequate substitutes for validated therapeutic benefit. Where AI tools have shown benefits in narrow studies, those findings cannot automatically be generalized to general purpose LLMs, different populations or acute crises [9].

## **8. Medicine as a measurable warning: the wrong prediction target**

Obermeyer and colleagues studied a widely used healthcare algorithm that relied on predicted medical expenditures as a proxy for patient need. Unequal patterns of spending meant that Black patients at a given risk score had substantially greater illness burden than White patients. The authors estimated that addressing this mismatch would raise the proportion of Black patients selected for extra care from 17.7 percent to 46.5 percent within the studied population [10]. The problem was not simply that statistical predictions could fail. The system was being asked to predict a proxy that reflected unequal care access rather than the underlying need that clinicians and administrators intended to identify.

This result should not be generalized to every medical AI model. Its scientific importance is to demonstrate a repeatable mechanism: choice of target variable can encode institutional inequities while standard performance indicators still look favorable. Analogously, gross active hour rideshare earnings may be measured accurately while failing to represent a worker's actual economic welfare. The relevant comparison is proxy selection and governance, not equivalent medical and transportation harms.

## **9. Why advancing AI can increase rather than reduce institutional exposure**

Greater predictive power may improve routing, scheduling and clinical analysis. Yet sophistication can also increase integration, complexity, dependence and the speed of propagation. When a recommendation becomes a default action and then part of a multi step autonomous workflow, small errors can cause downstream effects before a human notices. The risk depends on severity, reversibility, monitoring and scope of permission, not merely on model accuracy.

Model drift adds another difficulty. A system validated under one distribution of demand, prices, regulations or medical practices may perform differently after those conditions change. Organizations need version records, prospective validation, subgroup monitoring and defined conditions for disabling or revising systems. NIST frames this as a continuing governance task across its Govern, Map, Measure and Manage functions [4].

## 10. Proposed GRI research architecture

### Pillar A: observational Uber study

Create a structured longitudinal case record of public policy changes and contemporaneous field evidence. Distinguish documents from remembered experiences. Identify when offers were received, selected, declined or not observed; capture total online hours, pickup mileage, occupied mileage, tips, promotions, costs, neighborhood, vehicle, platform, time and weather where legitimately available. Scrub rider identifying information. Retain an immutable source log and codebook. Treat prior screenshots and recollections as heterogeneous evidence until quality assessed.

### Pillar B: comparative field trials

Prospectively define matched observation windows across times of day and market conditions. Where feasible, compare comparable trips within one driver and among consenting drivers, controlling for distance, duration, geography, app version, incentives and demand. Test sensitivity to missing declined offers. Use hierarchical models for driver and time effects, negative controls where possible, and natural experiments around independently verified pricing policy changes. Predefine the primary result and report effect sizes with uncertainty intervals. A practical requirement is to distinguish the driver selection process from any underlying offer generation process.

### Pillar C: agentic management governance trial

In the GRI Keys AI Connect setting, compare human operated workflows, AI recommendation requiring approval and narrowly scoped delegated action under a fixed budget and safety policy. Record each recommendation, human override, action, error, rollback, operational outcome and participant experience. Begin with low stakes tasks and require permission gates for any financially consequential or externally visible action. Predefine termination rules. The objective is to measure whether increasing autonomy changes productivity and the institution's ability to detect and correct mistakes.

### Pillar D: comparative healthcare and mental health analysis

Conduct structured literature synthesis and interviews with qualified professionals rather than unlicensed experimentation on vulnerable people. Examine the validity of proxies, model calibration, subgroup fairness, automation bias and human escalation. Studies involving patients or sensitive mental health data require suitable clinical partnerships, ethical review, security controls and jurisdiction specific compliance. GRI should not test autonomous mental health interventions directly on distressed individuals outside an approved research environment.

## 11. Falsifiable propositions and competing explanations

| Proposition                  | Possible evidence                                                     | Disconfirming or competing evidence                                   |
|------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| H1: earnings cluster         | Repeated distributions centered near a band across defined conditions | Differences explained by supply, selection and time windows           |
| H2: transparency gap         | Driver can observe quote but not reconstruct determining factors      | Complete reproducible rate rules available and sufficient             |
| H3: welfare divergence       | Active hour pay differs materially from online net return             | Differences small after consistent cost and time definitions          |
| H4: psychosocial association | Voluntary distress indicators covary with opacity and volatility      | Relationship attenuates after controls or reverses in matched samples |
| H5: autonomy risk            | More agent permissions increase undetected or irreversible mistakes   | Validation and controls hold error impacts constant or lower          |

The central aim is not to accumulate anecdotes confirming a conclusion. It is to test hypotheses against rival explanations and publish null findings where warranted. A GRI evidence grading scheme can distinguish verified primary evidence, published academic evidence, reproducible field measurement, self reported observation and theoretical conjecture.

## 12. A governance design for probabilistic authority

The proposed Probabilistic Authority Paradox can be expressed as a conceptual relationship: institutional exposure is a function of uncertainty, delegated autonomy, potential harm, observability and capacity for correction. It is not a validated risk equation and should not be presented with invented coefficients or a numerical institutional risk score. Organizational design should instead institute explicit controls suited to the stakes.

- Define objectives: document the legitimate human outcome as well as measurable proxies and excluded welfare costs.
- Validate probabilities: test calibration, subgroup outcomes, dataset representativeness and model drift against relevant ground truth.
- Limit permissions: distinguish informational outputs from recommendations, reversible actions and irreversible decisions.
- Require accountable ownership: designate a person or governing body with authority to investigate, override and remediate.
- Make decisions contestable: preserve meaningful explanations, human appeal routes and auditable records.

- Protect mental health: do not substitute automated ratings or conversational empathy for human assessment, accommodations or clinical expertise.
- Monitor realized consequences: review net worker economics, adverse subgroup outcomes and long term human welfare, not only efficiency dashboards.

This framework can align with NIST AI RMF without pretending that the same procedural threshold is appropriate for every domain. The tolerable error on a route recommendation is categorically different from a missed psychiatric emergency or denial of clinically necessary care.

## 13. Research limitations and disclosure

The paper offers a conceptual synthesis and research agenda. It does not report a completed randomized trial, access Uber proprietary source code, establish intentional individualized pay discrimination, or demonstrate a causal impact of platform management on psychiatric diagnoses. Field figures from GRI notes are provisional until documented with standardized source evidence. The sports prediction illustration is anecdotal and cannot establish performance without a prospectively captured sample. Mental health conclusions are grounded in institutional guidance and published research, not in clinical experimentation by GRI.

These limitations are part of the contribution. Institutional trust requires a research process that makes uncertainty visible instead of replacing it with persuasive language. A robust counterargument must also be preserved: algorithmic systems may reduce arbitrary human inconsistency, improve access to transport or clinical support, and provide useful early warnings. The test is whether their net benefits persist under meaningful welfare, fairness and accountability measures.

## 14. Conclusion

**The defining challenge of AI is not eliminating uncertainty. It is determining when uncertain computational outputs may legitimately become decisions that govern human work, opportunity and health.**

The progression from AI answering questions to AI managing people demands renewed attention to the human subject of management. In corporate America, workers have historically relied on policies, supervisors, professional norms and channels of appeal, all imperfect but institutionally identifiable. Algorithmic systems can strengthen these structures if leaders select defensible objectives, validate models and preserve human recourse. They can also weaken them when opaque outputs are treated as unchallengeable authority. Uber offers a tractable empirical setting, mental health exposes the cost of reducing people to signals, and medicine shows how accurate prediction of the wrong target can institutionalize inequity. These cases together form the foundation for a GRI comparative program on probabilistic authority and accountable management.

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